Practice workbook · Name receptors and the senses or stimuli they detect

Naming the structure that detects a stimulus, listing senses, naming skin temperature receptors, and reasoning about having several cone types.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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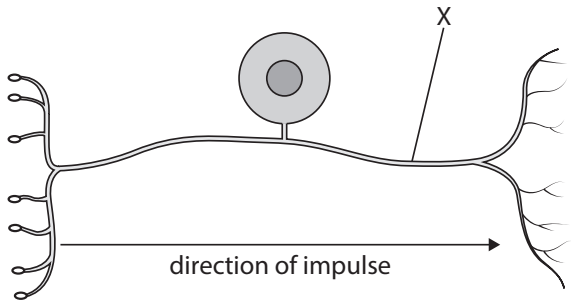
Type ID: HBI-NAME-RECEPTOR-AND-SENSE

1. Question 1 · part c.iii · 1 mark

June 2024 Paper 1

Worked-example retry · Attempt part c.iii only.

1 (a) The diagram shows a cell found in the human nervous system.



(Source: © Designua/ Shutterstock)

(i) The cell shown in the diagram transmits electrical impulses from a receptor cell to the central nervous system.

State the name of this type of nerve cell.

(1)

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(ii) Name the structure labelled X.

(1)

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1. Question 1 · part c.iii · 1 mark

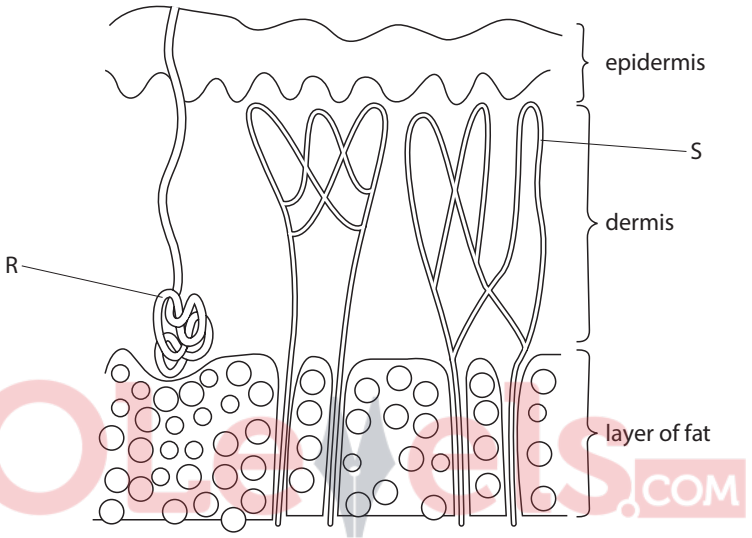
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Worked-example retry · Attempt part c.iii only.

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(b) Structures in the skin respond to information from the nervous system.

The diagram shows a section of the skin.



(Source: adapted from Human Physiology and Health, Author David Wright, Publisher Heinemann page 113)

(i) Name part R and part S.

(2)

part R

.....

part S

.....

(ii) Describe how part R helps to cool the body.

(2)

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1. Question 1 · part c.iii · 1 mark

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Worked-example retry · Attempt part c.iii only.

(iii) Describe the role of the layer of fat in maintaining body temperature.

(2)

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1. Question 1 · part c.iii · 1 mark

June 2024 Paper 1

Worked-example retry · Attempt part c.iii only.

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(c) A student investigates the effect of temperature on receptors in the skin.

This is their method.

- place one hand in hot water
- place the other hand in ice cold water
- after 2 minutes take both hands out of the water
- place both hands in water at room temperature

(i) State one possible hazard with their method.

(1)

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(ii) Describe how to reduce the risk of this hazard.

(2)

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(iii) When the student placed both hands in the water at room temperature the water felt hot to one hand and ice cold to the other.

Name the structure in the skin that detects temperature.

(1)

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(Total for Question 1 = 12 marks)

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1. Question 1 · part c.iii · 1 mark · Working and answer

Working and answer

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2. Question 4 · part a · 2 marks

June 2023 Paper 1

Further practice · Attempt part a only.

4 The eye contains receptors that can detect light.

(a) Name two other senses.

(2)

1 .....

2 .....

(b) The table shows how the focal length of the eye lens changes as the thickness of the lens changes.

Thickness of lens in mm	Focal length in mm
10	12.5
9	14.0
8	15.5
7	18.0
6	21.0
5	25.0

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(i) Describe the relationship between the thickness of the lens and its focal length.

(2)

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(ii) Give the effect on the diameter of the lens as its thickness increases.

(1)

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2. Question 4 · part a · 2 marks · Working and answer

Working and answer

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3. Question 7 · part a.i · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.i only.

7 (a) The nervous system sends electrical impulses to parts of the body in response to an external stimulus.

(i) Name the structure in the nervous system that detects an external stimulus.

(1)

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(ii) Name the structure that transmits electrical impulses directly to the central nervous system.

(1)

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(b) Neurones are specialised cells.

(i) Describe what is meant by a specialised cell.

(2)

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3. Question 7 · part a.i · 1 mark · Working and answer

Working and answer

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4. Question 4 · part c.iii · 2 marks

June 2023 Paper 1

Further practice · Attempt part c.iii only.

4 The eye contains receptors that can detect light.

(a) Name two other senses.

(2)

1 .....

2 .....

(b) The table shows how the focal length of the eye lens changes as the thickness of the lens changes.

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(i) Describe the relationship between the thickness of the lens and its focal length.

(2)

.....

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(ii) Give the effect on the diameter of the lens as its thickness increases.

(1)

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4. Question 4 · part c.iii · 2 marks

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Further practice · Attempt part c.iii only.

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(iii) A person is looking at a distant object and then looks at a near object.

Use the data in the table to explain the changes that occur in the eye so the person is able to form an image of the near object on the retina.

(4)

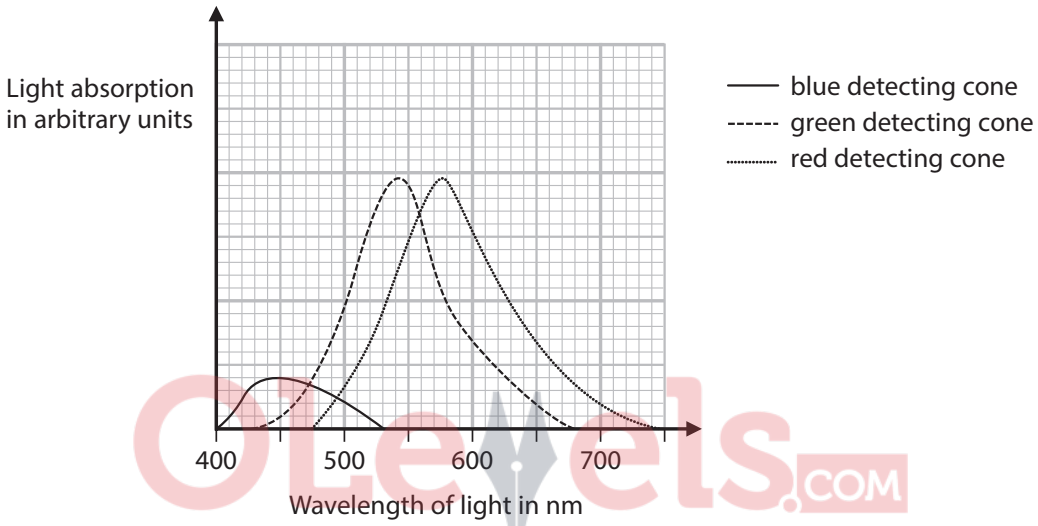
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4. Question 4 · part c.iii · 2 marks

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Further practice · Attempt part c.iii only.

(c) Cone cells are found in the retina of the eye. They detect colour. The graph shows the absorption of different wavelengths of light by three types of cone cell.



(i) Determine the wavelength of light at which a red detecting cone absorbs most light.

(1)

wavelength of light = ... nm

(ii) Determine the maximum difference in the wavelengths of the light that can be absorbed by blue detecting cones.

(2)

difference in wavelengths = ... nm

(iii) Suggest why it is better to have three types of cone rather than just one type of cone.

(2)

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(Total for Question 4 = 14 marks)

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4. Question 4 · part c.iii · 2 marks · Working and answer

Working and answer

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